# 1 Maps

Let $S$ and $T$ be two sets. A map $f$ from $S$ into $T$ is defined as a triplet $(S, T, G_f)$, where $G_f$ is a subset of $S \times T$ satisfying the following property:

$$\forall x \in S,\ \exists! y \in T : (x, y) \in G_f.$$

Intuitively, a map from $S$ into $T$ is a correspondence (or process) that associates each element of $S$ with a unique element $y$ in $T$. Here:

· The set $S$ is called the domain (or source) of $f$.

· The set $T$ is called the codomain (or target) of $f$.

· The element $y$ is called the image of $x$ under $f$ and is expressed as $y = f(x)$.

· The element $x$ is called the preimage or antecedent of $y$ under $f$.

· We can represent the map $f$ using the notation:

$$\begin{array}{rccc} f: & S & \longrightarrow & T \\ & x & \longmapsto & f(x). \end{array}$$

This notation indicates that $f$ maps elements from the set $S$ to elements in the set $T$, and for each $x \in S$, the image $f(x)$ is uniquely determined in $T$.

· We denote by $S^T$ the set of all maps from $S$ into $T$.

**Example 1**

1. *Let $S = \{a, b, c\}$, $T = \{1, 2, 3\}$, and let $G_f$, $G_g$ be the parts of $S \times T$ given by*
$$G_f = \{(a, 1), (b, 1), (a, 2), (c, 3)\}$$
*and*
$$G_g = \{(a, 2), (b, 1), (c, 3)\}.$$
*It is clear that $f$ is not a map but $g$ is a map from $S$ into $T$.*

2. *For any set $S$ the following map*
$$\begin{array}{rccc} id_S: & S & \longrightarrow & S \\ & x & \longmapsto & x \end{array}$$
*is called identity map of $S$.*

*3. Let $S$ and $T$ be two sets. For a fixed element $a$ in $T$. The map*

$$\begin{array}{rccc} f_a: & S & \longrightarrow & T \\ & x & \longmapsto & a \end{array}$$

*is called constant. This map sends every element of $S$ to the same element $a$ in $T$.*

*4. Let $S$ and $S^{'}$ be two sets such that $S^{'} \subset S$. The map*

$$\begin{array}{rccc} i_{S'}: & S^{'} & \longrightarrow & S \\ & x & \longmapsto & x \end{array}$$

*is called the canonical injection. This map embeds $S'$ into $S$ by mapping each element of $S'$ to itself in $S$.*

*5. Let $S$ and $S^{'}$ be two sets such that $S^{'} \subset S$. The map*

$$\begin{array}{rccl} \chi_{S'}: & S & \longrightarrow & \{0,1\} \\ & x & \longmapsto & \begin{cases} 1 \text{ if } x \in S^{'} \\ 0 \text{ otherwise,} \end{cases} \end{array}$$

*is called the indicator map of $S^{'}$.This map indicates membership in $S'$ by mapping elements to 1 if they are in $S'$ and to 0 otherwise.*

*6. Let $E$ and $F$ be two sets. The map*

$$\begin{array}{rccc} \rho_S: & S \times T & \longrightarrow & S \\ & (x,y) & \longmapsto & x \end{array}$$

*is called the projection map over $S$. In the same way the map*

$$\begin{array}{rccc} \rho_T: & S \times T & \longrightarrow & T \\ & (x,y) & \longmapsto & y \end{array}$$

*is called the projection map over $T$. These maps extract the first and second components, respectively, from the pairs in $S \times T$.*

*7. Let $f_1$ and $f_2$ be two maps defined as follows*

$$\begin{array}{rccc} f_1: & E_1 & \longrightarrow & T_1 \\ & x & \longmapsto & f_1(x) \end{array}$$

*and*

$$\begin{array}{rccc} f_2: & S_2 & \longrightarrow & T_2 \\ & y & \longmapsto & f_2(y). \end{array}$$

*The map product is defined as*

$$\begin{array}{rccc} f_1 \times f_2: & S_1 \times S_2 & \longrightarrow & T_1 \times T_2 \\ & (x,y) & \longmapsto & (f_1(x), f_2(y)). \end{array}$$

8. *A choice function on a set $S$ is any map $f$ from $\mathcal{P}(S)$ into $S$ such that for any $T \subset \mathcal{S}$ we have $f(T) \in T$.*
   *Consider the set $\mathbb{N}$. It is known that every subset has a smallest element. We can therefore define the function $f : \mathcal{P}(\mathbb{N}) \to \mathbb{N}$, which transforms any subset of $\mathbb{N}$ into its smallest element. This function satisfies: for every $X \in \mathcal{P}(\mathbb{N})$, we have $f(X) \in X$; in other words, it allows us to "choose" an element from each subset of the set $\mathbb{N}$. The map $f$ is a choice function on $\mathbb{N}$. Note that for a set like $\mathbb{Q}$, we do not explicitly know such a function, but we can demonstrate its existence using the axiom of choice.*

► One of the equivalent formulations of the choice axiom is

> For any collection of non-empty, pairwise disjoint subsets of a set $A$,
> there exists a subset of $A$ that intersects each of these subsets in exactly one element.

## 1.1 Equality of maps

Two maps

$$\begin{array}{rccc} f: & S & \longrightarrow & T \\ & x & \longmapsto & f(x), \end{array}$$

and

$$\begin{array}{rccc} g: & S' & \longrightarrow & T' \\ & x & \longmapsto & g(x) \end{array}$$

are said to be equal if $S = S'$, $T = T'$, and $\forall x \in S$: $f(x) = g(x)$.

**Example 2** *The following three maps are not equal*

$$\begin{array}{rccc} f: & \mathbb{R}^+ & \longrightarrow & \mathbb{R}^+ \\ & x & \longmapsto & x^2, \end{array}$$

$$\begin{array}{rccc} g: & \mathbb{R}^+ & \longrightarrow & \mathbb{R} \\ & x & \longmapsto & x^2, \end{array}$$

$$\begin{array}{rccc} h: & \mathbb{R} & \longrightarrow & \mathbb{R} \\ & x & \longmapsto & x^2, \end{array}$$

because, even the three formulas are equal, the sources and the targets are not equal. But there are relations between them, the following notions clarify this.

## 1.2 Operations on maps

Let

$$\begin{array}{rccc} f: & S & \longrightarrow & T \\ & x & \longmapsto & f(x) \end{array}$$

be a map, and let $A$, $G, B$ such that $A \subset S \subset G$, and $B \subset T$.

1. The restriction of $f$ to $A$ is the map denoted $f_{|A}$ defined by:

$$\begin{array}{rccc} f_{|A}: & A & \longrightarrow & T \\ & x & \longmapsto & f(x). \end{array}$$

2. An extension of $f$ to $G$ is any map $g : G \to T$ such that

$$\forall x \in S, g(x) = f(x).$$

3. The direct image of $A$ by $f$, denoted $f(A)$, is the subset of $T$, defined by:

$$f(A) = \{f(x), x \in A\} = \{y \in T, \exists x \in A : \ y = f(x)\}.$$

4. The preimage of $B$ by $f$, denoted $f^{-1}(B)$, is the subset of $S$ defined by:

$$f^{-1}(B) = \{x \in S : f(x) \in B\}.$$

**Example 3**

1. *Let $f$ and $h$ be the following maps:*

$$\begin{array}{rccc} f: & \mathbb{R} & \longrightarrow & \mathbb{R} \\ & x & \longmapsto & |x+1|, \end{array}$$

$$\begin{array}{rccc} h: & ]-1,\infty[ & \longrightarrow & \mathbb{R} \\ & x & \longmapsto & x+1. \end{array}$$

   *$h$ is the restriction of $f$ to $]-1,\infty[$.*

2. *Let $f$ and $g$ be the following maps:*

$$\begin{array}{rccc} f: & \mathbb{R}^+ & \longrightarrow & \mathbb{R} \\ & x & \longmapsto & x \end{array}$$

$$g: \quad \begin{array}{c} \mathbb{R} \to \mathbb{R} \\ x \longmapsto \begin{cases} x & \text{if } x \geq 0 \\ 1 & \text{if } x < 0. \end{cases} \end{array}$$

   *The map $g$ is an extension of $f$ to $\mathbb{R}$.*

3. *Let $f$ be the map defined as*

$$f: \begin{array}{ccc} \mathbb{R} & \longrightarrow & \mathbb{R} \\ x & \longmapsto & x^2. \end{array}$$

*We have* $f(\{-1,1,0\}) = \{0,1\}$ , $f(\mathbb{R}^-) = \mathbb{R}^+$, $f^{-1}([-2,2]) = \left[-\sqrt{2}, +\sqrt{2}\right]$ , $f^{-1}(]-\infty, 0[) = \emptyset$.

4. *Let $S$ be a set, $A$ a subset of $S$, and let $f$ be the map defined as*

$$f: \begin{array}{ccc} \mathcal{P}(S) & \longrightarrow & \mathcal{P}(S) \\ X & \longmapsto & X \cap A. \end{array}$$

*We have* $f(\{\overline{A}\}) = \{\emptyset\}$ , $f^{-1}(\{\emptyset\}) = P(\overline{A})$, $f(\{\overline{A}, A\}) = \{\emptyset, A\}$ .

**Proposition 4** *Let $f : E \to F$ be a map, $A$ and $A'$ two subsets of $E$, $B$ and $B'$ two subsets of $F$. Then we have*

1. $A \subseteq A' \Longrightarrow f(A) \subseteq f(A')$.
2. $f(A \cup A') = f(A) \cup f(A')$.
3. $f(A \cap A') \subseteq f(A) \cap f(A')$.
4. $B \subseteq B' \Longrightarrow f^{-1}(B) \subseteq f^{-1}(B')$.
5. $f^{-1}(B \cup B') = f^{-1}(B) \cup f^{-1}(B')$.
6. $f^{-1}(B \cap B') = f^{-1}(B) \cap f^{-1}(B')$.
7. $A \subseteq f^{-1}(f(A))$.
8. $f(f^{-1}(B)) \subseteq B$.
9. $f(A) \backslash f(A') \subseteq f(A \backslash A')$.
10. $f^{-1}(B \backslash B') = f^{-1}(B) \backslash f^{-1}(B')$.

**Proof.**

1. Let $A \subseteq A'$. Since $y \in f(A)$ is equivalent to $y = f(x)$ for some $x$ in $A$, and $A \subseteq A'$ we have $y = f(x)$ and $x \in A'$, then $y \in f(A')$.

2. We have $y \in f(A \cup A')$ is equivalent to $\exists x \in A \cup A' : y = f(x)$, $x \in A \cup A'$, which leads to $x \in A$ or $x \in A'$and therefore

$$f(x) \in f(A) \text{ or } f(x) \in f(A'),$$

which means that $y \in f(A) \cup f(A')$. Then

$$f(A \cup A') \subset f(A) \cup f(A') \tag{1}$$

Conversely, $A \subseteq A \cup A'$ gives $f(A) \subseteq f(A \cup A')$ and $A' \subseteq A \cup A'$ gives $f(A') \subseteq f(A \cup A')$. Then

$$f(A) \cup f(A') \subseteq f(A \cup A') \tag{2}$$

The inclusions (1) and (2) give the desired equality.

3. $y \in f(A \cap A')$ implies $y = f(x)$ with $x \in A \cap A'$. This leads to $y = f(x)$ with $x \in A$ and $x \in A'$. Then $y \in f(A)$ and $y \in f(A')$. Therefore

$$f(A \cap A') \subseteq f(A) \cap f(A').$$

4. Assume $B \subseteq B'$. For any $x \in E?x \in f^{-1}(B)$ is equivalent to $f(x) \in B$. Since $B \subseteq B'$, we have $f(x) \in B'$, equivalently $x \in f^{-1}(B')$. Thus,

$$f^{-1}(B) \subseteq f^{-1}(B').$$

5. For any $x \in F$, $x \in f^{-1}(B \cup B')$ is equivalent to $f(x) \in B \cup B'$, that is $f(x) \in B$ or $f(x) \in B'$ so $x \in f^{-1}(B)$ or $x \in f^{-1}(B')$, which is equivalent to

$$x \in (f^{-1}(B) \cup f^{-1}(B')).$$

This prroves that the equality in 5) holds.

6. It can be proven in the same way as 5).

7. Let $x \in A$. Then, $f(x) \in f(A)$ is equivalent to $x \in f^{-1}(f(A))$. proving

$$A \subseteq f^{-1}(f(A)).$$

8. In the same way as 7).

9. Let $y \in f(A) \backslash f(A^{'})$. Then $y \in f(A)$ and $y \notin f(A^{'})$. This gives that $y = f(x)$ with $x \in A$ and $x \notin A^{'}$, which implies $y = f(x)$ and $x \in A \backslash A'$. Therefore, $x \in f(A \backslash A')$ and so the inclusion

$$f(A) \backslash f(A') \subseteq f(A \backslash A').$$

10. If $x \in f^{-1}(B \backslash B')$, then $f(x) \in (B \backslash B')$, that is, $f(x) \in B$ and $f(x) \notin B'$, which is equivalent to $x \in f^{-1}(B)$ and $x \notin f^{-1}(B')$ i.e., $x \in f^{-1}(B) \backslash f^{-1}(B')$. Then the equality

$$f^{-1}(B \backslash B') = f^{-1}(B) \backslash f^{-1}(B').$$

■

Let $f, g, h$ be maps defined as follows

$$\begin{array}{ccc} & {}^{f}\rightarrow & \\ E & & F \\ h\nwarrow & G & \nearrow g \end{array}$$

The diagram is said to be commutative if $g = f \circ h$.
Let $f, g, h$, and $\varphi$ be maps defined as follows

$$\begin{array}{ccc} E & \overset{f}{\rightarrow} & F \\ \downarrow \varphi & & \uparrow g \\ G & \overset{h}{\rightarrow} & H \end{array}$$

The diagram is said to be commutative if $f = g \circ h \circ \varphi$.

## 1.3 Injection, surjection and bijection

**Definition 5** *Let $f : E \rightarrow F$ be a map.*

1. *$f$ is said to be injective or one to one if*

$$\forall x, x' \in E : \; f(x) = f(x') \Longrightarrow x = x'.$$

2. *$f$ is said to be surjective or onto if*

$$\forall y \in F, \exists x \in E : y = f(x).$$

3. *$f$ is said to be bijective if it is injective and surjective.*

The following characterizations of injection, surjection and bijection are easy to prove.

**Proposition 6** *Let $f : E \rightarrow F$ be a map.*

1. *$f$ is injective if and only if $\forall x', x'' \in E$: $x' \neq x'' \Rightarrow f(x') \neq f(x'')$.*

2. *$f$ is surjective if and only if $f(E) = F$.*

3. *$f$ is bijective if and only if for all $y \in F$ there is a unique $x \in E$ such that $y = f(x)$.*

**Example 7**

1. *Let $f$ be the map defined by:*

$$f: \begin{array}{ccc} \mathbb{R} & \longrightarrow & \mathbb{R} \\ x & \longmapsto & |x+1|. \end{array}$$

*We have $f(1) = f(-3)$ even though $+1 \neq -3$. So $f$ is not injective; nor surjective, since for $y < 0$ there is no real $x$ such that $f(x) = y$.*

2. *The map $f$ defined by:*

$$f: \begin{array}{ccc} \mathbb{N} & \longrightarrow & \mathbb{N} \\ k & \longmapsto & 2k \end{array}$$

*is injective but not surjective.*

▶ Note that if a map

$$f: \begin{array}{ccc} E & \longrightarrow & F \\ x & \longmapsto & f(x) \end{array}$$

is not surjective, we can canonically deduce a surjection by "reducing" the codomain to the image ($f(E)$). Note that to demonstrate surjectivity, we can either compute the image or show that every element of the codomain is the image of at least one element. Conversely, to demonstrate non-surjectivity, it suffices to find an element in the codomain that has an empty preimage.

## 1.4 Composition and inverse maps

Let $f$ and $g$ be the maps defined by:

$$f: \begin{array}{ccc} E & \longrightarrow & F \\ x & \longmapsto & f(x), \end{array}$$

$$g: \begin{array}{ccc} F & \longrightarrow & G \\ x & \longmapsto & g(x). \end{array}$$

The composition of $f$ and $g$ taken in this order, and denoted $g \circ f$, is the map defined by

$$g \circ f: \begin{array}{ccc} E & \longrightarrow & G \\ x & \longmapsto & g(f(x)). \end{array}$$

**Example 8**

1. *Let $f$ and $g$ be the following maps defined as:*

$$f: \begin{array}{ccc} \mathbb{N} & \longrightarrow & \mathbb{N}, \\ n & \longmapsto & 2n \end{array}$$

*and*

$$\begin{array}{rcl} g: \mathbb{N} & \longrightarrow & \mathbb{N} \\ n & \longmapsto & \begin{cases} \frac{n}{2} \text{ if } n \text{ is even,} \\ 0 \text{ otherwise.} \end{cases} \end{array}$$

*We have* $g \circ f = id_{\mathbb{N}}$ *and*

$$\begin{array}{rcl} f \circ g: \mathbb{N} & \longrightarrow & \mathbb{N} \\ n & \longmapsto & \begin{cases} n \text{ if } n \text{ is even} \\ 0 \text{ otherwise.} \end{cases} \end{array}$$

*This example allows us to say that, in general,* $f \circ g \neq g \circ f$.

2. *Let:*

$$\begin{array}{rcl} f: \mathbb{R} & \longrightarrow & \mathbb{R} \\ x & \longmapsto & x^2. \end{array}$$

*and*

$$\begin{array}{rcl} g: \mathbb{R}^+ & \longrightarrow & \mathbb{R} \\ x & \longmapsto & \sqrt{x}. \end{array}$$

*We can compose* $f$ *and* $g$ *since* $\operatorname{Im} f = \mathbb{R}^+ \subset \mathbb{R}^+$.
*We have*

$$\begin{aligned} (g \circ f)(x) &= g(f(x)) \\ &= g(x^2) \\ &= \sqrt{x^2} \\ &= |x|. \end{aligned}$$

*The composed map is therefore*

$$\begin{array}{rcl} g \circ f: \mathbb{R} & \longrightarrow & \mathbb{R} \\ x & \longmapsto & |x|. \end{array}$$

*We can also compose* $g$ *and* $f$ *since* $\operatorname{Im} g = \mathbb{R}^+ \subset \mathbb{R}$,
*We have*

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= f(\sqrt{x}) \\ &= (\sqrt{x})^2 \\ &= x. \end{aligned}$$

*We find here*

$$\begin{array}{rcl} f \circ g: \mathbb{R}^+ & \longrightarrow & \mathbb{R} \\ x & \longmapsto & x. \end{array}$$

**Definition 9** *Let* $f : S \to T$ *be a bijective map. The map denoted by* $f^{-1}$, *and defined from* $T$ *into* $S$ *which associates to any* $y \in T$ *the unique element* $x \in S$ *such that* $y = f(x)$, *is called the inverse map of* $f$.

$$\begin{array}{rccc} f^{-1}: & T & \longrightarrow & S \\ & x & \longmapsto & f^{-1}(x). \end{array}$$

Thus, we have:

$$f^{-1}(y) = x \Leftrightarrow y = f(x),$$

where $x \in S$ and $y \in T$.

**Remark 10**

**a)** *Let $S$ and $T$ be two sets. The map $h$ defined by*

$$\begin{array}{rccc} h: & Bij(S,T) & \longrightarrow & Bij(T,S) \\ & f & \longmapsto & f^{-1} \end{array}$$

*is bijective, where $Bij(S,T)$ denotes the set of bijections from $S$ into $T$.*

**b)** *Let $f$ be a bijective map from $S$ into $T$. Note that we have*

$$S \xrightarrow{f} T \xrightarrow{f^{-1}} S$$

*so*

$$x \xrightarrow{f} y = f(x) \xrightarrow{f^{-1}} f^{-1}(y) = x,$$

*therefore*

$$(\forall x \in S) \left(f^{-1} \circ f\right)(x) = x$$

*that is, $f^{-1} \circ f$ is the identity map on $S$ i.e., $f^{-1} \circ f = id_S$. In a similar way, we have $f \circ f^{-1} = id_T$.*

**c)** *Let $f$ be a map from $S$ into $T$. For any $y \in T$ we have*

$$f^{-1}(\{y\}) = \{x \in S : f(x) = y\}.$$

*Then $f^{-1}(\{y\})$ is a subset of $S$ and is defined even $f$ is not bijective. But $f^{-1}(y)$ is an element of $S$ and is defined only when $f$ is bijective and in this case we have*

$$f^{-1}(\{y\}) = \left\{f^{-1}(y)\right\}.$$

**Example 11** *The map*

$$\begin{array}{rccc} f: & \mathbb{R} & \longrightarrow & \mathbb{R} \\ & x & \longmapsto & x^2. \end{array}$$

*is neither injective nor surjective, but the map*

$$\begin{array}{rccc} g: & \mathbb{R}^+ & \longrightarrow & \mathbb{R}^+ \\ & x & \longmapsto & x^2. \end{array}$$

*is bijective, as is the map*

$$\begin{array}{rccc} h: & \mathbb{R}^- & \longrightarrow & \mathbb{R}^+ \\ & x & \longmapsto & x^2. \end{array}$$

*The inverse bijection of $g$ is the map*

$$\begin{array}{rccc} g^{-1}: & \mathbb{R}^+ & \longrightarrow & \mathbb{R}^+ \\ & x & \longmapsto & \sqrt{x}, \end{array}$$

*and the inverse of $h$ is the map*

$$\begin{array}{rccc} h^{-1}: & \mathbb{R}^+ & \longrightarrow & \mathbb{R}^- \\ & x & \longmapsto & -\sqrt{x}. \end{array}$$

**Proposition 12** *Let $f : E \to F$ and $g : F \to G$ be two maps.*

1. *If $f$ and $g$ are injective, then $g \circ f$ is injective.*
2. *If $f$ and $g$ are surjective, then $g \circ f$ is surjective.*
3. *If $f$ and $g$ are bijective then, $g \circ f$ is bijective and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.*

**Proof.**

1. For any $x, x' \in E$ : we have $(g \circ f)(x) = (g \circ f)(x')$ is equivalent to $g(f(x)) = g(f(x'))$. This gives $f(x) = f(x')$ because $g$ is injective. Then $x = x'$ because $f$ is injective. Therefore $g \circ f$ is injective.

2. Let $z \in G$. Since $g$ is surjective, there exists $y \in F$ such that $z = g(y)$. On the other hand, $f$ is surjective, so there exists $x \in E$ such that $y = f(x)$, which gives $z = g(y) = g(f(x)) = (g \circ f)(x)$. Therefore, $g \circ f$ is surjective.

3. The bijectivity of $g \circ f$ follows from (1) and (2). On the other hand, for all $z \in G$. We have

   $$(g \circ f)^{-1}(z) = x \Longleftrightarrow z = (g \circ f)(x)$$

   but we have

   $$\left(f^{-1} \circ g^{-1}\right)(z) = f^{-1}\left(g^{-1}(z)\right) = f^{-1}(y)$$

   (with $z = g(y) = x$ and $y = f(x)$), where

   $$\forall z \in G : \left(f^{-1} \circ g^{-1}\right)(z) = (g \circ f)^{-1}(z),$$

   which means that

   $$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

■

**Proposition 13** *Let $S$ and $T$ be two sets, and let $f : S \to T$ and $g : T \to S$ be two functions such that $g \circ f = id_S$. Then $f$ is injective and $g$ is surjective.*

**Proof.** Let $x, x'$ be elements of $S$ such that $f(x) = f(x')$. Then composing both sides with $g$ we have $g(f(x) = g(f(x'))$, which means $(g \circ f)(x) = (g \circ f)(x^{'})$. Since $g \circ f = id_S$, we get $x = x'$; thus, $f$ is injective. Now let $v$ be an element of $S$. Since $g \circ f = id_S$ we have $v = g(f(v))$, so $v$ is the image under $g$ of an element in $T$ (namely $f(v)$), which means $g$ is surjective. ■

**Proposition 14** *Let $S$ and $T$ be two sets, and let $f : S \to T$ and $g : S \to T$ be two maps such that*

$$g \circ f = id_S \tag{3}$$

*and*

$$f \circ g = id_T. \tag{4}$$

*Then*

$$f,\ g \text{ are bijective, and } g = f^{-1}.$$

**Proof.** By (3), $f$ is injective, $g$ is surjective, and by (4), $g$ is injective, and $f$ is surjective. Therefore, $f$ and $g$ are bijective. Moreover, if $y$ is an element of $T$, the equality $f(g(y)) = y$ is equivalent to $g(y) = f^{-1}(y)$, thus $g = f^{-1}$. Similarly, we see that $f = g^{-1}$. ■

## 1.5 Sequences and Indexed sets:

- Let $S$ be a set. A sequence of $n$ elements from $S$ is obtained by assigning an element $s_i$ from $S$ for any $i = 1, 2, \ldots, n$. A sequence of $n$ elements from $S$ is therefore nothing other than a map from $\{1, 2, \ldots, n\}$ into $S$. If $f$ denotes this map. The values of $f$ are usually denoted as $(s_1, s_2, \ldots, s_n)$ or $(s_i)_{1 \leq i \leq n}$.

- Similarly, we define an infinite sequence of elements from $S$. It is a map from the set $\{1, 2, 3, \ldots\}$ into $S$. Such a sequence is denoted as $(s_1, s_2, \ldots, s_n, \ldots)$ or $(s_i)_{i \geq 1}$.

- Let $S$ be a set and $I$ an auxiliary set that we will call the index set. A family of elements from $S$, indexed by $I$, is by definition a mapping from $I$ to $S$. Such a family is generally denoted by $(s_i)_{i \in I}$.

- A family of sets $(S_i)_{i \in I}$ is a family such that each $S_i$ is a set. For such a family, we can define a union and an intersection, which generalize the notions introduced for a finite number of sets. The union of the $S_i$, denoted $\bigcup_{i \in I} S_i$, is the set of elements that belong to at least one $S_i$. The intersection of the $S_i$, denoted $\bigcap_{i \in I} S_i$, is the set of objects that belong to all the $S_i$.

## 1.6 Equipotence and countability

- Two sets are said to be equipotent when there is a bijection from one to the other.

**Example 15**

1. *Let $S$ be a set. The map*

$$h: \begin{array}{ll} \mathcal{P}(S) & \longrightarrow S^{\{0,1\}} \\ X & \longmapsto \chi_X \end{array}$$

*is bijective, where $\chi_X$ is the indicator map of $X$. Then $\mathcal{P}(S)$ is equipotent to $S^{\{0,1\}}$.*

2. *The sets $\mathbb{N}$ and $\mathbb{Z}$ are equipotent. We define a map $f : \mathbb{N} \to \mathbb{Z}$ as follows:*

$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even,} \\ -\frac{n+1}{2} & \text{if } n \text{ is odd.} \end{cases}$$

- A set $S$ is said to be countable if there exists a bijection between $S$ and a subset of $\mathbb{N}$.
- A set is uncountable if it is not countable.

**Example 16** *Finite sets are countable sets.*

**Proposition 17** *If $S$ is both countable and infinite, then there is a bijection between $S$ and $\mathbb{N}$.*

**Example 18** *The set of integers $\mathbb{Z}$ and the set of rational numbers $\mathbb{Q}$ are infinite and countable.*

**Proposition 19** *Let $S$ and $S'$ be two sets.*

1. *If $S$ is countable and $S' \subset S$, then $S'$ is also countable.*
2. *If $S' \subset S$ and $S'$ is uncountable, then $S$ is also uncountable.*

**Proof.**

1. Since $S$ is countable, there is a bijection $f : S \to \mathbb{N}$. Consequeltly, if we let $f(S') = T$ then $T$ is a subset of $\mathbb{N}$, and $f$ induces a bijection between $S'$ and $T$.
2. This is the contrapositive of Assertion 1).

■